

Agnos Candidate Assignment

Each candidate will need to complete the assignment within 3 days of receiving the task.

DevOps:

Tasks:

Overview:

Design and implement a production-ready DevOps setup. The focus is on reliability, security, CI/CD, and observability. Application logic can be minimal or stubbed.

1. System Components
 - a. API Service, A simple API service
 - i. can be any language/framework
 - ii. Required API endpoints:
 1. GET /health , return status of the service
 - b. Background Service Worker, A simple background service worker
 - i. can be any language/framework
 - ii. Required worker job:
 1. Update timestamp of today records
2. Docker Containerization
 - a. Multi-stage Docker builds
 - b. Configuration via environment variables
 - i. Each environment have their own environment variables (DEV , UAT, PROD)
3. Kubernetes
 - a. Design to have high availability as much as possible
 - b. Separate Deployments for API and Worker
 - c. Resource requests and limits
 - d. Readiness and liveness probes
 - e. Horizontal Pod Autoscaler for the API
4. CI/CD
 - a. Implement a CI/CD pipeline using GitHub Actions, or equivalent.
 - b. Pipeline stages:
 - i. Lint and tests
 - ii. Build container images
 - iii. Security scan (Optional)
 - iv. Deploy step (real or mocked)
5. Monitoring
 - a. Structured JSON logs

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- b. Metrics: request latency, error rate
 - c. Alerts (e.g. high error rate, stalled worker, crash looping)
- 6. Failure Scenarios (short answer of what to do in these scenarios)
 - a. API crashes during peak hours
 - b. Worker fail and infinitely retries
 - c. Bad deployment is released
 - d. Kubernetes node downs
- 7. Tech Stack
 - a. Docker
 - b. Kubernetes
 - c. CI/CD: GitHub Actions or equivalent
 - d. Monitoring: Prometheus or equivalent
 - e. Hosting: Local or cloud (optional)

Deliverables:

Share your work to khochapak@gmail.com and hr@agnoshealth.com

- 1. Git repository containing:
 - a. Dockerfiles
 - b. Kubernetes manifests
 - c. CI/CD configuration
- 2. README including:
 - a. Architecture overview
 - b. Setup instructions
 - c. Usage instructions
 - d. Failure Scenario handling documents

Evaluation Criteria:

- 1. **Reliability and failure handling**
- 2. **Kubernetes fundamentals**
- 3. **Observability quality**
- 4. **Security**
- 5. **Clarity of documentation**